k = 4 index calculus on $E(\mathbb{F}_{p^4})$, measured

relations $\propto p^{1.02}$, Wiedemann $\propto p^{1.93}$, rho $\propto p^{1.86}$; crossover from the fits 2^{101} ($h = 4$), interval 2^{91}–2^{117}

`research/gaudry_quartic_d4_exponent_20261015` · 2026-10-15 to 2026-10-27 · class: measurement

Summary. Gaudry/Diem index calculus with FGHR's D_4 oracle (PR #1863) and rho with the negation map were run end to end in ecbench on `fp4_torsion` curves for 12 primes $p = 29 \ldots 1009$: 146 sessions, 740 records, all verified ($[d]G = Q$), all audited (674 replays reproduced exactly), `S` a count. The plan had three parts: a ladder of 72 jobs (one curve per rung), 12 of which ran after the first stop at the owner's request, and an extension (amendment 5, frozen with predictions before any extension run) with two more curves per rung up to $p = 509$, large primes at higher p, and third targets at the top rungs. Fitted with curves as a level of the design, the relation phase grows like $p^{1.02}$ (95 %: 0.84–1.19), the linear algebra (Wiedemann) like $p^{1.93}$ (1.86–2.02), and rho like $p^{1.86}$ (1.77–1.94) (q^1, q^2, q^2 as the analysis says). Index calculus still loses to rho at every measured size ($7.2 \cdot 10^5\times$ at $p = 29$, $6.7 \cdot 10^4\times$ at $p = 1009$). From the fits it meets rho at a group of $2^{101.4}$ ($h = 4$; cluster bootstrap $2^{95.1}$–$2^{115.9}$; with the relations slope at its interval ends $2^{90.6}$–$2^{116.8}$), against the §11.16 model's $2^{105.4}$–$2^{106.5}$, which lies inside both. **This extrapolates 15.4 bits in p, 61.5 bits in group size, beyond the top rung.** At that point the relation phase is 99 % of the cost. Large primes cut the matrix 3–5 times but cost 1.1–6.4× more on paired walks, the penalty growing with p, and push the fitted crossover out (2^{145} for two large primes). The 72-job fit, made before the extension, gave $2^{97.9}$ and a wider interval (2^{82}–2^{127}); both are reported. All ten extension predictions held.

1. Method

Protocol `PROTOCOL.md`, registered at `08722a2b7` before any session; amendments: (1) queue reordered and stop extended to 16 h (runtime, before any IC record); (2) audit scope by size; (3) relaunch of 20 jobs after the container's CPU set shrank; (4) the 12 jobs that missed the stop run at the owner's request, with a corrected count; (5) the extension below, with (5a) and (5b) two repairs after the container lost CPUs and was restarted. Code `0bea75038`. Arms (all `fghr-y-line:zero=1`, oracle `gaudry-d4:macaulay=26,weighted=1,chains=1,sparsest=1,replay=1,charge_unsolved=1`): `ic-d4-w` (`linalg=wiedemann`), `ic-d4-lp2`/`ic-d4-lp1` (`large_primes=2`/`1`, small base the first $\lceil N^\theta \rceil$ columns, $\theta = \frac{3}{4}, \frac{7}{9}$), `rho-neg`. One spec per (arm, p, target); the IC arms on a target share the walk. 60 of 72 jobs ran before the 16 h stop; the other 12 (third plain targets at five small rungs, second LP2 targets at seven) were run afterwards at the owner's request (amendment 4, same specs and binary). Earlier versions of this report said 59 and 13: that was a miscount.

New code. `linalg=wiedemann` solves for the target through a left-kernel vector of the relation matrix, $O(m \cdot \mathrm{nnz} + m^2)$ multiplications mod r, priced as `row_ops`; the default dense elimination ran on the same rows as an unpriced shadow and **gave the same logarithm in all 60 IC runs**. The large-prime wrapper works on any summand oracle; the existing large-prime code is binary-field only.

Fits. OLS of $\ln(\text{rung mean})$ on $\ln p$; interval = union of the Student-t 95 % interval and a bootstrap (10,000, targets resampled within rungs). The crossover solves $\sum_k A_k p^{\beta_k} = \overline{S}_{\mathrm{rho}} \sqrt{p^4/h}$.

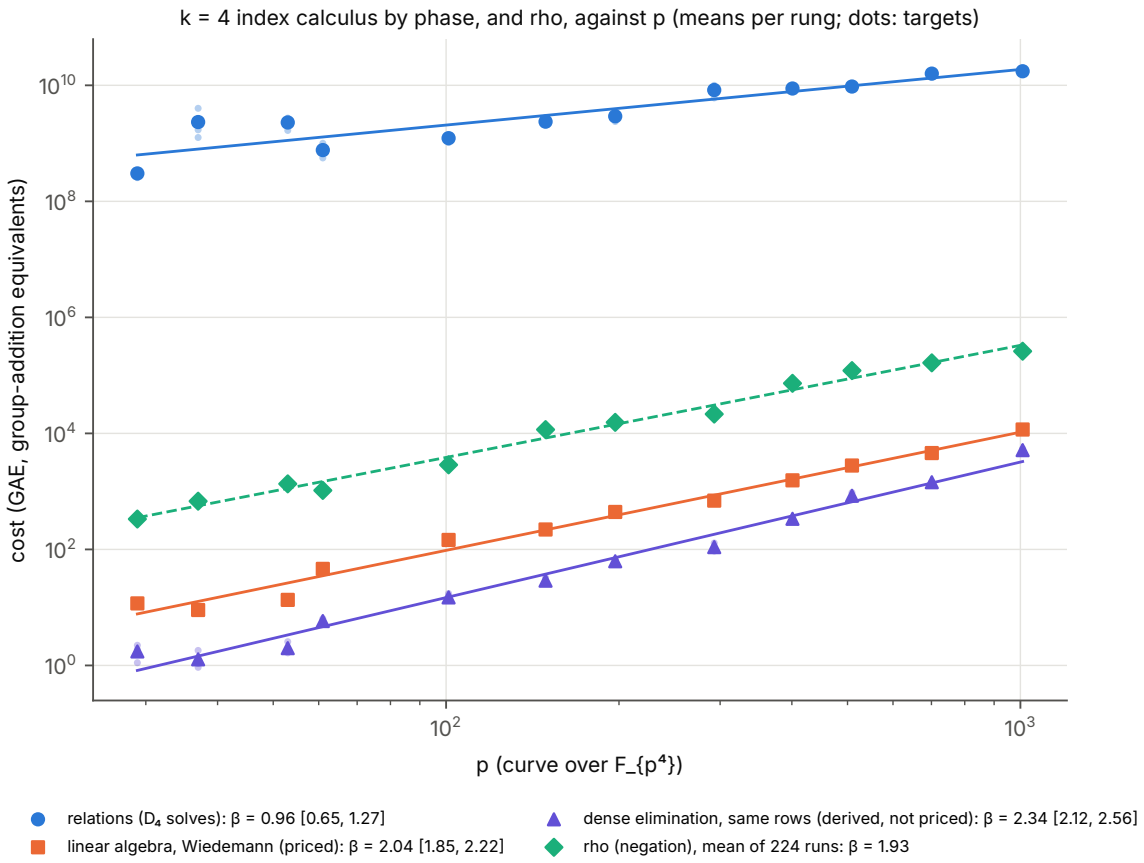


Figure 1: Cost by phase and rho against p (rung means; light dots: single targets). The dense elimination is derived from its `row_ops` at the same price and is not part of `S`.

2. Results (the first 72 jobs)

quantity	β	**95 % interval**	**expected**
relations phase (= total)	**0.96**	0.65–1.27 (bootstrap 0.90–1.03)	q^1
trials	0.85	0.54–1.17	
linear algebra, Wiedemann	**2.04**	1.85–2.22	q^2
dense elimination, same rows (derived)	2.34	2.12–2.56	q^3 worst case
rho (224 distinct runs)	**1.93**	1.78–2.09	q^2

The relations interval is wide because trials per relation vary between curves (rung mean trials over mean relations: 56 at $p = 29$, 75 at 61, 383 at 53, 496 at 37, 81 at 1009), which the within-rung bootstrap does not see. One D_4 solve costs $6.1 \cdot 10^5$ GAE at $p = 29$ and $8.3 \cdot 10^5$ at $p = 1009$; 171 of 220,652 systems (0.08 %) were degenerate and are charged. $\overline{S}_{\mathrm{rho}} = 1.08$ (1.02–1.15), drifting from 1.6 at $p = 29$ to 0.9–1.0 at the top.

crossover (group size)	$h = 2$	$h = 4$	$h = 16$
measured fits, point	$2^{96.0}$	$\mathbf{2^{97.9}}$ ($p^* = 2^{24.5}$)	$2^{102.0}$
bootstrap within rungs	$2^{92.8}$–$2^{100.4}$	$2^{94.6}$–$2^{102.5}$	$2^{98.4}$–$2^{106.8}$
relations slope at 0.65 / 1.27	$2^{80.6}$–$2^{124.6}$	$\mathbf{2^{82.1}}$–$\mathbf{2^{127.4}}$	$2^{85.2}$–$2^{133.2}$
§11.16 model (#1863)	$2^{103.0}$–$2^{103.7}$	$2^{105.4}$–$2^{106.5}$	—

With rho's own fitted power law in p: $2^{105.4}$ ($2^{99.9}$–$2^{113.7}$). The top rung is a $2^{39.9}$ group, so the crossover is **58 bits of group size (14.5 bits of p) beyond the measurements**, 7.5–8.6 bits of group size earlier than the model. The honest interval contains the model's figure; the bootstrap interval does not see between-curve scatter and understates the uncertainty. With the dense elimination priced instead, the cost meets rho at $2^{98.9}$ ($h = 4$) and rises above it again at 2^{128}; at $h = 16$ it meets it at $2^{106.3}$ and is above again from 2^{114}. With Wiedemann the fitted cost stays below rho through the extrapolated range ($p \le 2^{34}$).

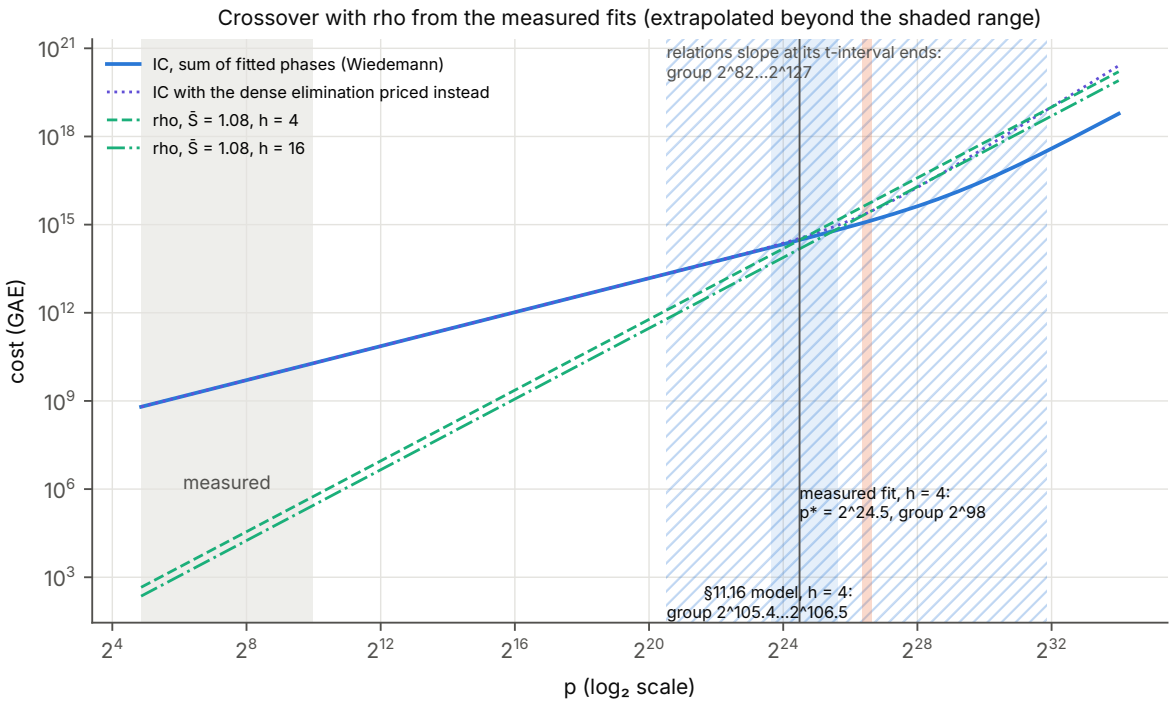


Figure 2: Extrapolation of the fitted phases to the crossover with rho. Grey: the measured range. Blue band: bootstrap; hatched: relations slope at its interval ends; orange: the §11.16 model at $h = 4$.

3. Large primes (the first 72 jobs)

arm	runs (rungs)	lin. alg. β	relations β	crossover $h = 4$
none	32 (12)	2.04 (1.85–2.22)	0.96 (0.65–1.27)	$2^{97.9}$
two, $\theta = \frac{3}{4}$	16 (8, $p \le 293$)	**1.70** (1.25–2.15)	1.26 (0.65–1.86)	$2^{127.7}$
one, $\theta = \frac{7}{9}$	12 (6, $p \le 149$)	1.62 (1.11–2.14)	1.49 (0.43–2.55)	2^{178} (153–214)

Paired S ratios (same target and walk), two large primes, two targets per rung: 1.00/1.00, 1.00/1.00, 0.91/1.12, 1.09/1.36, 1.09/1.32, 1.22/1.80, 1.80/2.31, 1.44/2.28 for $p = 29 \ldots 293$; one: 0.89–1.12 at $p \le 53$, 1.78–3.45 at $p \ge 61$. The matrix shrinks as the analysis says (Wiedemann 3.1–4.7× cheaper at $p = 293$) and the fitted slope falls from 2.04 to 1.70, but the intervals (1.25–2.15, 1.85–2.22) overlap, so it is consistent with the theory's 1.5 and with 2.04. The matrix is 10^{-6} of the cost here and 3 % at the crossover; the extra trials land on the relations phase. Large primes would pay only where the linear algebra dominates, past about 2^{117}. With the relations slope at 1.86 the double-large-prime cost never meets rho below $p = 2^{64}$.

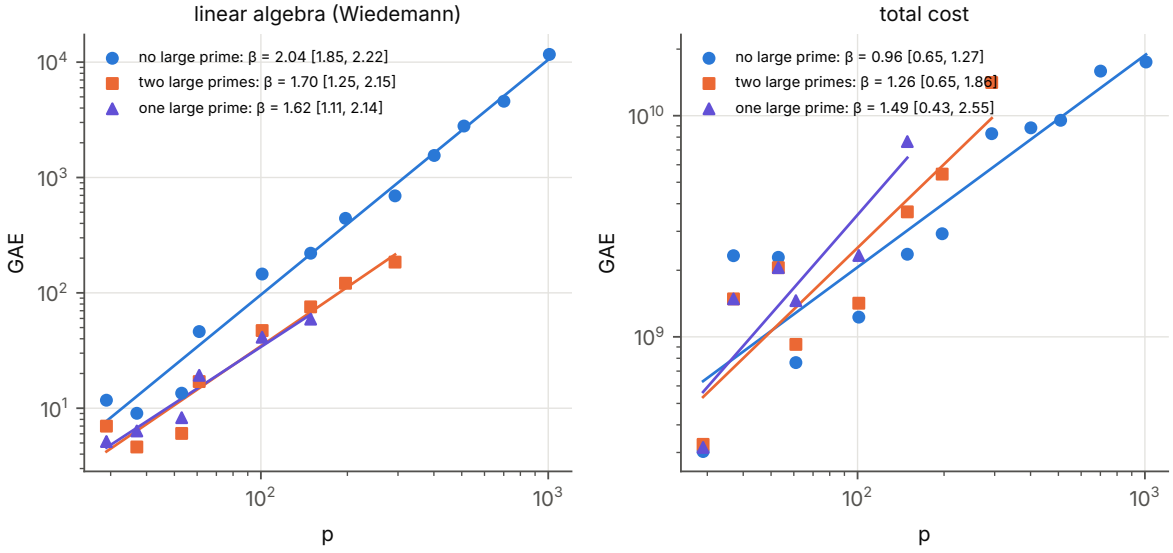


Figure 3: Linear algebra and total cost with and without large primes.

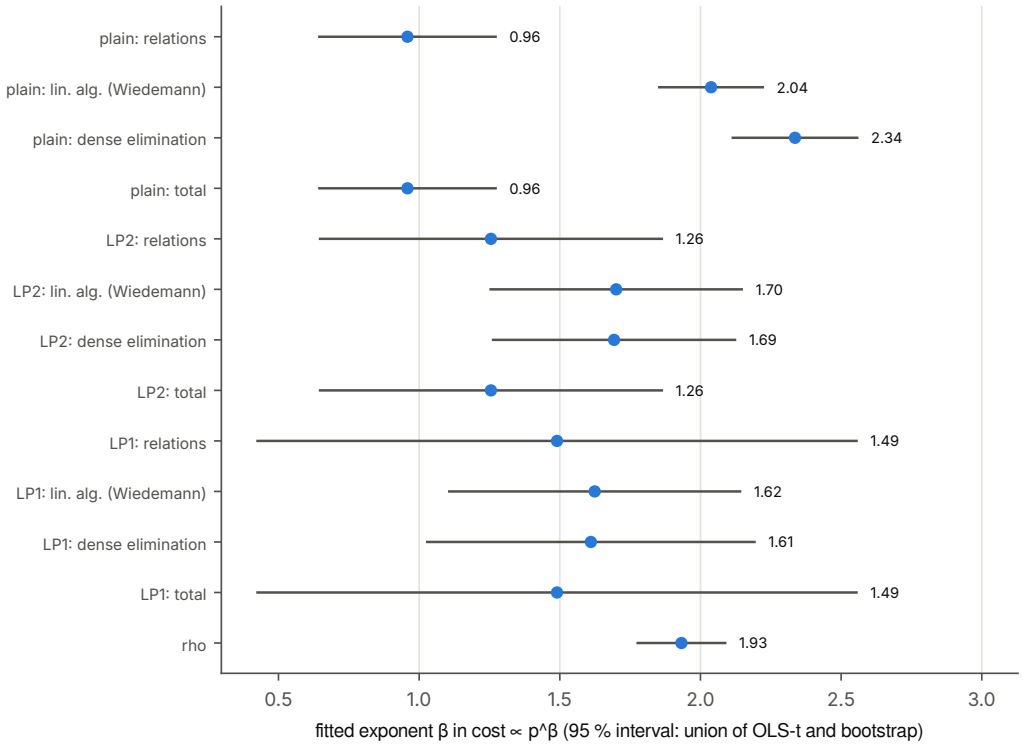


Figure 4: All fitted exponents with their 95 % intervals.

4. Last 12 jobs: the fit before and after

The 60-job fit (`raw/summary-60-jobs.json`, commit `9fb5c60e6`) was computed and read before the decision to run the remaining 12; it is kept. The 72-job fit is the headline because it uses the full planned set.

quantity	**60 jobs**	**72 jobs**
relations β (plain)	1.00 (0.74–1.26)	0.96 (0.65–1.27)
Wiedemann β	2.04 (1.85–2.22)	2.04 (1.85–2.22)
rho GAE β	1.93 (1.77–2.08)	1.93 (1.78–2.09)
LP2 linear algebra β	1.57 (1.12–2.01)	1.70 (1.25–2.15)
crossover $h = 4$ (point; bootstrap; slope band)	$2^{100.4}$; $2^{98.0}$–$2^{103.6}$; $2^{85.9}$–$2^{124.9}$	$2^{97.9}$; $2^{94.6}$–$2^{102.5}$; $2^{82.1}$–$2^{127.4}$
crossover, LP2, $h = 4$	$2^{123.9}$	$2^{127.7}$

5. The extension (amendment 5)

What ran. 74 sessions, all completed: the plain arm on two more curves at each of $p = 29…509$ (40; the curves come from chaining start seeds, `curves.tsv`, because `fp4_torsion(p, seed, 16)` returns the first curve at or after the seed and seeds 1–5 mostly give the same curve), 16 rho targets on each of those 20 curves, `lp2` at $p = 401, 509, 701$ and `lp1` at $p = 197, 293$ on the original curves and targets (10), and a third plain target at $p = 401, 509, 701, 1009$ (4). The container was restarted three times and lost CPUs; the resumable runner reran what was in flight (counts unaffected).

Fits (a rung is the mean over its curves of each curve's mean over its targets; interval = union of the Student-t and a cluster bootstrap over curves, then targets; $p \le 509$: three curves per rung, $p = 701, 1009$: one):

quantity	**72 jobs**	**extended**
relations (= total), plain	0.96 (0.65–1.27)	**1.02 (0.84–1.19)**
linear algebra, Wiedemann	2.04 (1.85–2.22)	**1.93 (1.86–2.02)**
dense elimination (derived)	2.34 (2.12–2.56)	2.24 (2.05–2.43)
rho GAE	1.93 (1.78–2.09)	**1.86 (1.77–1.94)**
`lp2` linear algebra / relations	1.70 / 1.26	1.72 (1.50–1.93) / 1.36 (1.06–1.66)
`lp1` linear algebra / relations	1.62 / 1.49	1.56 (1.32–1.81) / 1.64 (1.15–2.14)

The relations interval is 0.35 wide against 0.62.

crossover (group size)	$h = 2$	$h = 4$	$h = 16$
extended fit, point	$2^{99.4}$	$\mathbf{2^{101.4}}$ ($p^* = 2^{25.4}$)	$2^{105.5}$
cluster bootstrap	$2^{93.2}$–$2^{113.4}$	$2^{95.1}$–$2^{115.9}$	$2^{98.9}$–$2^{120.8}$
relations slope at 0.84 / 1.19	$2^{88.9}$–$2^{114.3}$	$\mathbf{2^{90.6}}$–$\mathbf{2^{116.8}}$	$2^{94.1}$–$2^{121.8}$
72-job fit, point; slope band	$2^{96.0}$; $2^{80.6}$–$2^{124.6}$	$2^{97.9}$; $2^{82.1}$–$2^{127.4}$	$2^{102.0}$; $2^{85.2}$–$2^{133.2}$
§11.16 model (#1863)	$2^{103.0}$–$2^{103.7}$	$2^{105.4}$–$2^{106.5}$	–

The top rung is a $2^{39.9}$ group: the crossover is **61.5 bits of group size (15.4 bits of p) beyond the measurements**, 4.0–5.1 bits earlier than the model, which lies inside the cluster-bootstrap interval and the slope band. The interval skews upward (a larger relations slope moves the crossover far out). With rho's own fitted power law: $2^{117.9}$ ($2^{107.9}$–$2^{142.7}$). The relation phase is 99 % of the cost at the crossover; the linear algebra overtakes it only past $p \approx 2^{32.8}$ (group $\approx 2^{131}$). With the dense elimination priced instead, the cost meets rho at $2^{101.8}$ and is above it again from $p = 2^{41.5}$ (group 2^{166}).

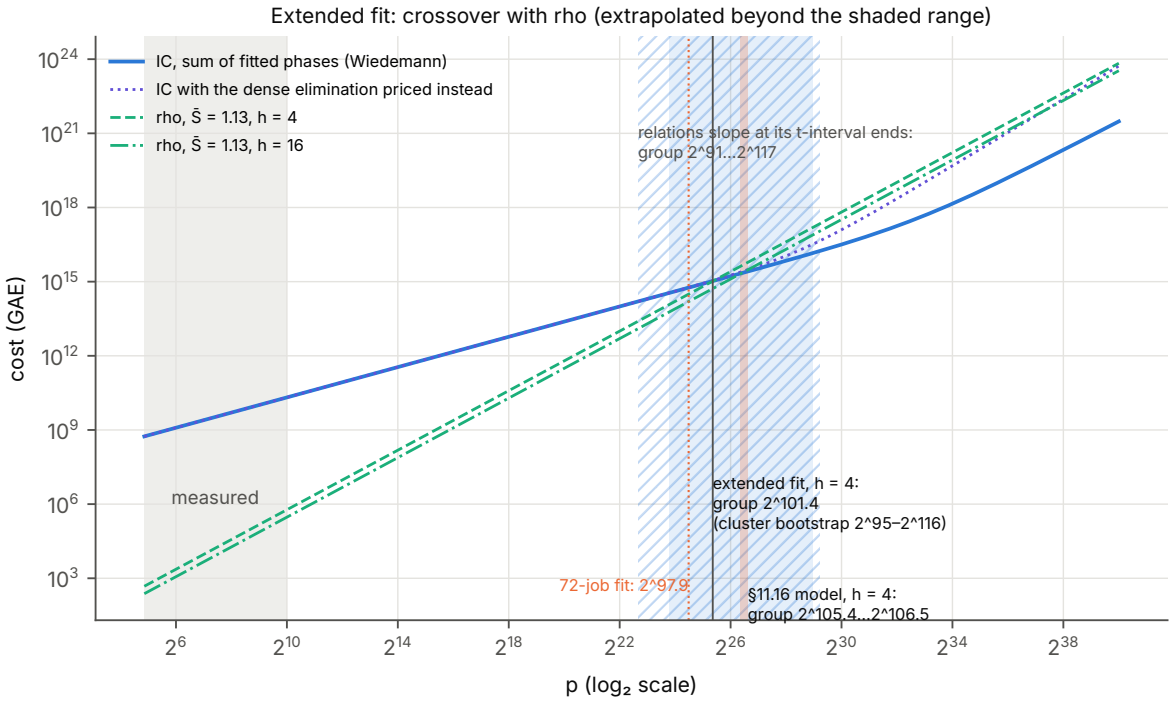


Figure 5: Extended fit: the fitted phases extrapolated to the crossover with rho. Grey: the measured range. Blue band: cluster bootstrap; hatched: relations slope at its interval ends; dotted orange: the 72-job fit; orange band: the §11.16 model at $h = 4$.

The scatter between curves. $\ln(\text{trials per relation})$ varies between curves with a pooled standard deviation of 0.73, but the spread falls with p: a factor of 14 at $p = 37$, 16 at $p = 53$, 5 at $p = 61$, and 1.2–3.1 from $p = 149$ up. The relations slope fitted on each chain of curves alone is 0.96 (0.65–1.28), 1.14 (0.78–1.51) and 1.23 (0.85–1.62), pooled 1.02. **Limit:** $p = 701$ and 1009 have one curve each, so the scatter between curves is not sampled at the two rungs with the most leverage on the slope.

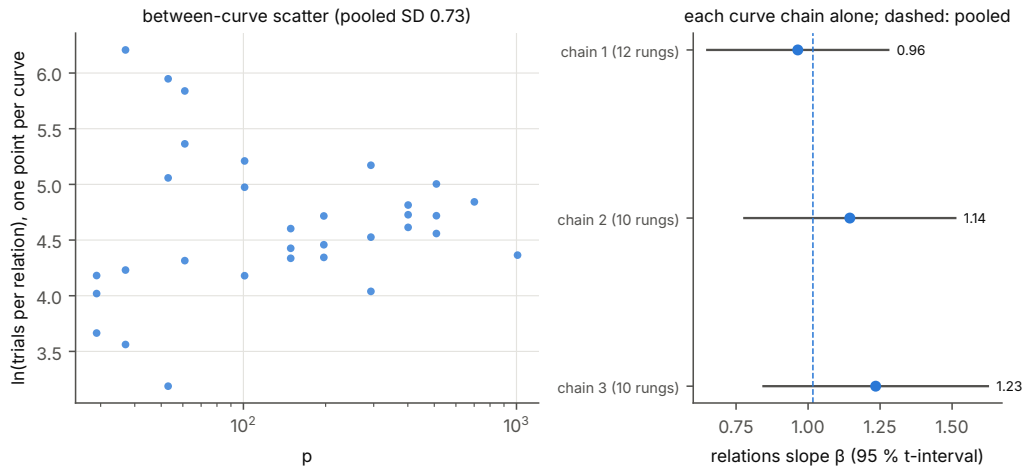


Figure 6: Left: $\ln(\text{trials per relation})$ per curve. Right: the relations slope on each curve chain alone, with its 95 % t-interval; dashed, the pooled fit.

Large primes at higher p (paired on the same walk; S over the plain arm's): `lp2` at $p = 197, 293, 401, 509, 701$: 1.80/2.31, 1.44/2.28, 2.13/2.58, 2.36/2.38, 2.88/2.91; `lp1` at $p = 197, 293$: 3.21/6.36, 2.37/5.15. The linear algebra is 3.1–4.8× cheaper, but the extra trials cost more than it saves and the penalty grows with p. Fitted crossovers: $2^{145.1}$ (2^{139}–2^{152}) for `lp2`, 2^{245} for `lp1` (3,591 of 10,000 replicates do not cross).

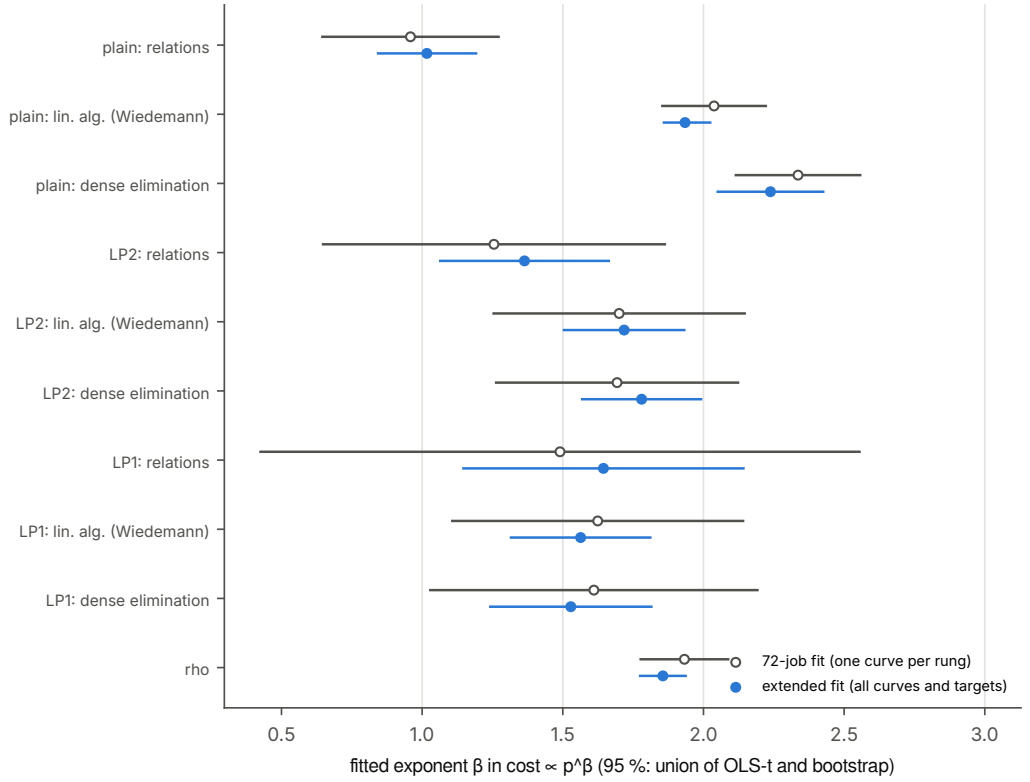


Figure 7: Fitted exponents with 95 % intervals: the 72-job fit (open) and the extended fit (filled).

Predictions X1–X10: all held. X1 740/740 verified, 674/674 replays reproduced; X2 relations slope 1.02 in $[0.75, 1.25]$; X3 t-interval width 0.35 < 0.50; X4 crossover $2^{101.4}$ in $[2^{88}, 2^{108}]$, slope band 26.2 bits < 38; X5 chain slopes 0.96/1.14/1.23 in $[0.5, 1.5]$; X6 pooled SD 0.73 in $[0.2, 1.0]$; X7 `lp2` ratios 2.13–2.91 at $p = 401, 509, 701$; X8 `lp1` ratios 2.37–6.36 at $p = 197, 293$; X9 `lp2` linear-algebra slope 1.72 in $[1.3, 2.0]$ and its crossover ($2^{145.1}$) later than the plain arm's; X10 rho slope 1.86 in $[1.6, 2.4]$.

6. Predictions (first 72 jobs)

P1 held: 312/312 verified, 290/290 replays reproduced. P2 held (0.96 ∈ [0.9, 1.3]). P3 held (2.04). **P4 not held** (dense elimination 2.34 < 2.4). P5 held (1.93; S_{rho} slope −0.15). P6: crossover $2^{97.9} \in [2^{88}, 2^{115}]$ held; **"no dense crossover below 2^{200}" not held** ($2^{98.9}$, recrossing at 2^{128}). P7 held (1.70; 1.26 > 0.96; every paired ratio at $p \geq 101$ above 1: 1.09–2.31). P8 held.

7. Audit

`ecbench verify` on all 72 sessions (receipts `audits/<job>.audit.json`; scope fixed by amendment 2): **audit OK on all 72, 312 records, 0 problems.** Full replay (`--replay-all`) on 61 sessions, every session with $p \leq 197$, every rho-only session and the spot checks `main-p293-t0` and `main-p401-t0`: **290 replays reproduced exactly**. Integrity only (stored $[d]G = Q$ checks, every planned run recorded, hashes) on the other 11: `main-p1009-t0/t1`, `main-p701-t0/t1`, `main-p509-t0/t1`, `main-p401-t1`, `main-p293-t1/t2`, `lp2-p293-t0`, `lp2-p293-t1`. The first 60 sessions were audited on 2026-10-16 and the last 12 on 2026-10-19, with the same `ecbench` binary.

Extension. `ecbench verify` on all 146 sessions: **audit OK on all 146, 740 records, 0 problems**. Full replay on 113 (every session with $p \leq 197$, every rho-only session, `xmain-c2-p293-t0` and `lp2-p401-t0`): **674 replays reproduced exactly**; integrity only on 33.

8. Limits

The exponents are fitted over $p = 29 \ldots 1009$ (groups 2^{19}–2^{40}); the crossover is an extrapolation of those fits, not a measurement, and its width is set by the scatter between curves in the relation rate and by the leverage of the top rungs, which have one curve each. The large-prime fits rest on two targets per rung over $p \leq 509$ (`lp2`) and $p \leq 293$ (`lp1`). S is a count on one host class; wall time is not admitted. No `docs/bounds` domain exists for $\mathbb{F}_{p^4}$ curves, so there is no challenge verdict. Study files: `README.md`, `raw/summary-ext.json`, `raw/summary.json`, `sessions/`, run reports `reports/run-20261015T0457Z/`, `reports/run-20261019T1612Z/` and `reports/run-20261020T0605Z/`.